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2.21 Photosynthesis

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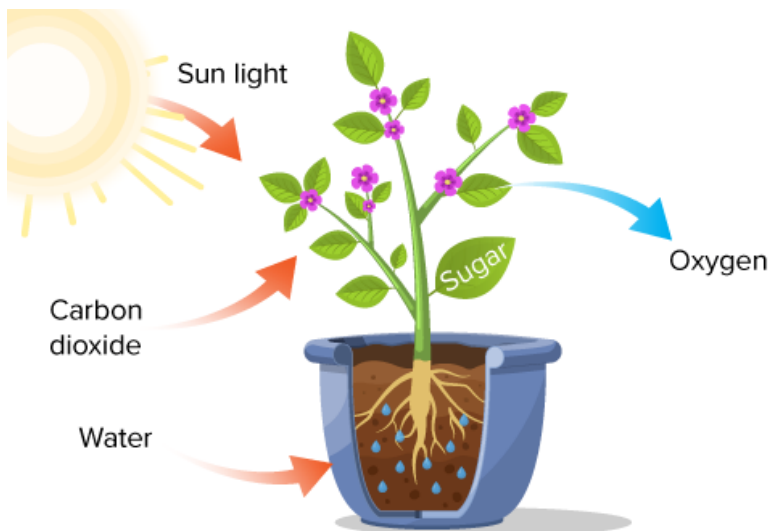
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What will you learn

- The process of photosynthesis
- How **chloroplasts** capture sunlight
- How light **energy** is converted into chemical energy



[Figure 1]

Oxygen has been described as a “waste product.” Is it?

Essentially, oxygen is a waste product of the **light reactions** of photosynthesis. It is a “leftover” from a necessary part of the process. All the oxygen that is necessary to maintain most forms of life just happens.

Photosynthesis Stage 1: The Light Reactions

Chloroplasts Capture Sunlight

Every second, the sun fuses over 600 million tons of hydrogen into approximately 596 million tons of helium, converting over 4 million tons of matter (4.3 billion kg) into light and heat energy. Countless tiny packets of that light energy travel 93 million miles (150 million km) through space, and about 1% of the light which reaches the Earth's surface participates in photosynthesis. Light is the source of energy for photosynthesis, and the first set of reactions which begin the process requires light – thus the name, **light reactions**, or light-dependent reactions.

When light strikes **chlorophyll** (or an accessory pigment) within the chloroplast, it energizes electrons within that molecule. These electrons jump up to higher energy levels; they have absorbed or captured, and now carry, that energy. High-energy electrons are “excited.” Who wouldn't be excited to hold the energy for life?

The excited electrons leave chlorophyll to participate in further reactions, leaving the chlorophyll “at a loss”; eventually they must be replaced. That replacement process also requires light, working with an **enzyme** complex to split **water** molecules. In this process of **photolysis** (“splitting by light”), H₂O molecules are broken into hydrogen **ions**, electrons, and oxygen atoms. The electrons replace those originally lost from chlorophyll. Hydrogen ions and the high-energy electrons from chlorophyll will carry on the energy transformation after the light reactions are over.

DID YOU KNOW?

Elysia chlorotica, a **species** of green sea slug, takes chloroplasts from **algae** and places them into its own digestive **cells**. This enables these **animals** to capture solar energy and perform photosynthesis.

The oxygen atoms, however, form oxygen gas, which is a waste product of photosynthesis. The oxygen given off supplies most of the oxygen in our **atmosphere**. Before photosynthesis evolved, Earth's atmosphere lacked oxygen altogether, and this highly reactive gas was toxic to the many **organisms** living at the time. Something had to change! Most contemporary organisms rely on oxygen for efficient **respiration**. So plants don't just “restore” the air, they also had a major role in creating it!

To summarize, chloroplasts “capture” sunlight energy in two ways. Light “excites” electrons in pigment molecules, and light provides the energy to split water molecules, providing more electrons as well as hydrogen ions.

Light Energy to Chemical Energy

Excited electrons that have absorbed light energy are unstable. However, the highly organized **electron carrier** molecules embedded in chloroplast membranes order the flow of these electrons, directing them through **electron transport chains** (ETCs). At each transfer, small amounts of energy released by the electrons are captured and put to work or stored. Some is also lost as heat with each transfer, but overall the light reactions are extremely efficient at capturing light energy and transforming it into chemical energy.

Sequential transport chains harvest the energy of excited electrons.

electrochemical gradient – a higher concentration of both positive charge and hydrogen inside the thylakoid than outside. The gradient formed by the H^+ ions is known as a **chemiosmotic gradient**. Picture this energy buildup of H^+ as a dam holding back a waterfall. Like water flowing through a hole in the dam, hydrogen ions “slide down” their **concentration gradient** through a **membrane protein** which acts as both **ion channel** and enzyme. As they flow, the ion channel/enzyme **ATP synthase** uses their energy to chemically bond a phosphate group to ADP, making **ATP**.

(2) Light re-energizes the electrons, and they travel down a second **electron transport chain (ETC)**, eventually bonding hydrogen ions to $NADP^+$ to form a more stable energy storage molecule, **NADPH**. NADPH is sometimes called “hot hydrogen,” and its energy and hydrogen atoms will be used to help build sugar in the second stage of photosynthesis.

NADPH and ATP molecules now store the energy from excited electrons – energy which was originally sunlight – in **chemical bonds**. Thus chloroplasts, with their orderly arrangement of pigments, enzymes, and electron transport chains, transform light energy into chemical energy. The first stage of photosynthesis – light-dependent reactions or simply light reactions – is complete.



The Light Reactions of Photosynthesis

RicochetScience



Watch on



in molecules of NADPH and ATP.

- The light reactions also release oxygen gas as a waste product.

Review

1. Summarize what happens during the light reactions of photosynthesis.
2. What is the chemiosmotic gradient?
3. Explain the role of the first electron transport chain in the formation of ATP during the light reactions of photosynthesis.



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